Notes on the MATLAB package implementing Classical AMG

The main folder contains, in addition to these notes, the two subfolders:

- **sources**: a folder that contains all the matlab and c++ source files (for generating the mex modules);
- **example**: a folder that contains a small structural problem to check the correct functioning of the code;
- **utilities**: a folder that contains a couple of script for generating matlab files from ascii input and generate 3D rigid body modes from coordinates.

Below follows a detailed description of each folder.

Sources

This folder is in turn divided into 7 subfolders, each containing specific parts of the code:

- **driver**: contains the main solver file that takes care of reading matrix, test space, right-hand side and solver parameters. It calls the AMG set-up and solves the related system using the MATLAB implementation of PCG or BiCGstab depending on whether the symmetry of the operator has been maintained or not;
- **aspAMG**: main function for the computation and application of the AMG;
- **smoother**: main driver for smoother computation. There is a subfolder "**c++**" that contains the c/c++ sources to generate the MEX module of FSAI. To generate the module, you have to run from MATLAB the script "compile.m" and then copy (or link) the file .mexa64 in the parent folder;
- **tspace**: main driver for calculating/improving the test space, if needed;
- **coarsen**: main driver for coarsening operations. There are four ways to create the Strength of Connection (**SoC**), affinity-based "***AFF***" (useful for test spaces composed by multiple vectors, e.g. elasticity), classical symmetrized "***CLA***", classical non-symmetrized "***CLANSY***" and diagonal dominance-based "***DOM***". The affinity based SoC uses ***SoC_perc*** as a filtering parameter, i.e., the percentage of connections to be kept with respect to the number of connections in the original matrix. Other SoCs simply use the tolerance ***tau***;
- **prolongation**: main driver for prolongation computation and its eventual smoothing. The implemented prolongations (in MATLAB or MEX version) are: Extended + I (useful for "Poisson-like" problems, e.g. CFD) and BAMG (more effective on structural problems). The parameters for EXTI are ***exti_type*** (1 or 2) to determine how to redistribute the error on the interpolatory set (BoomerAMG uses the approach 1) and distance (1 or 2) to determine whether to extend the interpolatory set to distance 1 or 2 (with 1 it looks like classical prolongation, with 2 it is actually extended prolongation). The parameters for the BAMG are very complicated and it is better not to touch them apart from ***prol_smooth*** that

determines how many smoothing steps to perform on the prolongation (in general 0 for CFD problems and 1 for mechanical problems). Finally, the parameter ***np*** determines for all MEX implementations the number of threads to use in the run (if the machine is multicore it is better to use ***np*** equal to the number of cores to speed up the calculation);

- **filter**: sources for prolongation and operator filtering. The filtering operations are available both in MATLAB and MEX. Prolongation filtering never causes problems, operator filtering causes a non-symmetry in the matrices of the next levels and therefore it is not 100% safe (sometimes if you filter too much you are unable to calculate the smoother FSAI). The parameters to control the filtering of the prolongation are ***filt_wgt*** ($\geq$ 0.0) that determines the relative weight in percentage of the filtered prolongation compared to the original one $\left\|P_f\right\|_F \sim (filt_{wgt}/100)\|P\|_F$ (obviously if ***filt_wgt*** $\geq$ 100, nothing is filtered). The recommended value is between 70.0 and 100.0. ***filt_tol*** is a tolerance used to avoid sharp jumps in the prolongation coefficients, the lower it is the less high the jumps. ***filt_tau*** (between 0 and 1) is a tolerance that determines the relative weight of the part that is deleted of the operator compared to the original operator $\left\|A - A_f\right\|_F \sim filt\_tau\|A\|_F$. If a value of ***filt_tau*** greater than zero is used, BiCGstab must be used as the solver because the preconditioner then becomes non-symmetric. Small values are recommended to avoid problems, e.g., between 0.0 and 0.1. ***min_patt*** safety parameter. If ***min_patt*** > 0 a minimum pattern is generated for the operator and the coefficients within that pattern are kept even below the filtering tolerance. Finally, the ***np*** parameter determines for all MEX implementations the number of threads to be used in the calculation;
- **utils**: some functions that are called in other parts of the code.

Example

This folder contains some input files to run a simple test case.

- ADDPATH.m is a script that add to the matlab path the folders containing source files. ADDPATH contains relative addresses, so if you don't move the folder, the program should work as is;
- aspAMG.fnames is an ASCII file containing the names of all the other files read by the driver. More specifically:
    - AMG parameters;
    - SMOOTHER parameters;
    - TEST SPACE parameters;
    - COARSENING parameters;
    - PROLONGATION parameters;
    - FILTERING parameters;
    - GENERAL parameters;
    - Name of the matrix in ASCII format (insert name even if not needed)
    - Name of the test space file in ASCII format (insert name even if not needed)

- Name of the right-hand side file in ASCII format (insert name if not needed)
- Name of the file in binary format (insert name if not needed)

All of the AMG parameters are commented in the main drivers of each code part. The GENERAL parameters are instead the following:

- **ascii_input**: if true it assumes reading in ascii format, if false in binary format;
- **rhs_build**: creation mode of the right-hand side:
  - 'unit_sol': right-hand side corresponding to unit solution;
  - 'unit_rhs': unitary right-hand side;
  - 'rand_sol': right-hand side corresponding to a random solution;
  - 'rand_rhs': random right-hand side;
  - 'rhs_in': right-hand side acquired in input;
- **solv_method**: solution method, PCG, BiCGstab or stationary AMG;
- **itmax**: maximum number of iterations allowed for;
- **tol**: output tolerance on the relative residual.

Utilities

The folder contains two tools, "Prepare_Binaries.m" which reads matrix, right-hand side, and initial test space in ascii format and prints them into a single MATLAB-specific binary .mat file (reading binary files in MATLAB is much faster), and "mk_rbm_3d.m" to generate the six rigid motion vectors (RBM) from the nodal coordinates of a model.